

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. (IT) Examination****March 2020****IT443 Language Processors****Date: 16.03.2020, Monday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Answer the following questions. [07]**
- 1) Define Language Processor. List out different Language Processor. [02]
 - 2) Compare NFA and DFA. [02]
 - 3) What is Symbol table? How symbol table differs from other data structure? [03]
- Q - 2.a Discuss phases of the compiler in detail. [06]**
- Q - 2.b Answer any TWO questions. [08]**
- 1) Define macro. Discuss different types of macro with example.
 - 2) Explain assembly language statements with example.
 - 3) Write regular expression and construct DFA for the given language of all strings $\Sigma \in \{a, b\}^*$ that end with ab.
- Q - 3 Answer any TWO questions. [14]**
- 1) Explain two pass assembly in detail.
 - 2) What is compiler? Explain compilation process in detail with phases of compiler.
 - 3) Discuss parser optimization and linear optimization techniques in detail.

SECTION – II

- Q - 4 Do as directed. [07]**
- 1) List down the different bottom-up parsing techniques in compiler. [01]
 - 2) Differentiate between compiler and interpreter. [02]
 - 3) Explain how lexical analysis is different than syntax analysis phase? [02]
 - 4) Define the terms: 1) Handle 2) Lexeme [02]
- Q - 5.a What is an ambiguous grammar? Explain one example of ambiguous grammar. [04]**

Q - 5.b Attempt any TWO questions.**[10]**

- 1) Draw model of LR parser and explain it.
- 2) Which phase of the compiler will generate Three Address Code? Write Three Address code for following code.

$$x = a * b + c * d - e * f$$

- 3) Write down the rules to find the First () and Follow () set of any grammar.

Q - 6.a List and explain the conflicts in shift-reduce parsing technique.**[04]****Q - 6.b Attempt any TWO questions.****[10]**

- 1) Write an advantages and limitation of operator precedence parsing technique.
- 2) List and explain the different types of top down parsing techniques in brief.
- 3) Consider the following grammar and Apply the brute force approach to trace the input string: accd.

Grammar: $S \rightarrow aAd/aB$

$A \rightarrow b/c$

$B \rightarrow ccd/ddc$

- 4) Explain the Non Recursive Predictive Parsing Algorithm in detail.
